



PROFESSIONAL SUMMARY

Entry-level AI & Full-Stack Developer holding an MCA and B.Sc. in Computer Science, with hands-on experience building full-stack, AI-powered, and blockchain applications using Java, Python, React, Node.js, Flask, and LLM APIs (Groq). Independently designed and deployed 8+ end-to-end projects, backed by certifications in SQL, Data Analysis, Prompt Engineering, and Anthropic's Claude/MCP tools. Seeking Software Developer / Full-Stack Developer / AI Engineer roles, available immediately with flexibility to relocate anywhere in India.

EDUCATION

MCA | Avanathi Institute of Engineering & Technology
2024 – 2026 | Status: Completed
Hyderabad

B.Sc. Computer Science | Kakatiya University
Completed 2024 | Warangal | CGPA: 6.51/10

Intermediate & 10th | Sri Nalanda Jr. College & Singareni Collieries High School
CGPA: 6.32/10 & 6.7/10 | Kothagudem

EXPERIENCE

Academic & Freelance Project Experience

Independent Software & AI Developer 2024 – Present

- Architected and deployed 8+ applications from scratch. The work spanned full-stack development, AI integration, and blockchain projects, and covered database design, REST APIs, and security
- Used AI-assisted development workflows (Claude, Groq) to move faster through prototyping and debugging, and to speed up deployment

TECHNICAL SKILLS

Languages: Java (OOP, Java 8+), Python, JavaScript, SQL, PL/SQL, C++

AI & LLM: Groq API, LLM Integration (LLaMA), Prompt Engineering, Cloud API Integration

Web Development: React.js (Vite), Node.js, Express.js, Flask, JSP, Servlets, HTML5, CSS3, REST API Design

Database: MySQL, PostgreSQL, CRUD Operations, Database Design

Architecture & Tools: Software Architecture, Fault-Tolerant System Design, Apache Tomcat, Git, GitHub, Docker, Postman

Security & Concepts: Counting Bloom Filters (CBF), Cryptographic Verification, Data Security & Encryption, Public Verifiability

CERTIFICATIONS

- SQL and Relational Databases 101
- Data Analysis Using Python
- Prompt Engineering for Everyone
- Deloitte:** Cyber Security Simulation, Data Analytics Simulation
- Anthropic:** Claude 101, Claude Code 101, Claude Code in Action, Claude with the Anthropic API, Introduction to MCP, MCP Advanced Topics
- Udemy:** Complete Prompt Engineering for AI Bootcamp (2026), Complete Full-Stack Web Development Bootcamp

KEY STRENGTHS

- AI-Assisted Efficiency:** uses modern AI tools (Claude, Groq, Prompt Engineering) to speed up prototyping and software development
- Independently conceived and built 8+ end-to-end projects, then deployed each one, while completing MCA coursework
- Strong problem-solving skills demonstrated through debugging deployment and API integration issues
- Quick learner. Self-taught new technologies (Groq API, Motoko/ICP, RAG) through independent research
- Strong focus on writing code that stays clean and scalable, and is easy to maintain

PROJECTS

AI Projects

Panda AI - Multi-User AI Assistant Platform

Flask, Python, Groq API, REST APIs

Wanted a personal AI assistant with real web search and file understanding, not just a text chatbot, so I architected this instead of depending on a single paid API.

- Engineered a multi-user AI assistant with real-time search grounding and multimodal file processing (PDF, image, audio, video)
- Designed multi-provider fallback architecture (3 Groq API keys plus OpenRouter) achieving zero-cost, high-availability deployment
- Architected stateless REST API design to support concurrent multi-user sessions without performance degradation

Universal AI-Powered Resource & Budget ERP System

Java, Groq LLM, REST API

Wanted to explore how AI could go beyond chatbots and actually help with real business decisions, like knowing when to reorder stock or flagging a suspicious transaction.

- Delivered a multi-domain ERP system with AI engine for business insights, auto-reordering, and fraud detection
- Developed REST API and browser dashboard supporting concurrent multi-user sessions across 4+ business modules

Java Projects

Secure Cloud Migration & Verifiable Deletion System

Java 8, Counting Bloom Filters, Cryptographic Verification, MySQL, Apache Tomcat

MCA Major Project (Avanathi Institute). Engineered this to solve the critical cloud security challenge of safely migrating data between providers and guaranteeing permanent, verifiable data deletion without relying on a third party.

- Engineered a cloud-to-cloud data transfer and verifiable deletion framework using Counting Bloom Filters (CBF), eliminating reliance on Trusted Third Parties (TTP)
- Implemented cryptographic data confidentiality, end-to-end integrity validation, and public verifiability protocols to detect malicious server operations

Inventory Management System

Java, Java Swing, ArrayList, OOP

Developed this to understand how businesses actually track profit and stock in real time, and to get hands-on with Java Swing for the first time.

- Created a GUI-based system using Java Swing with real-time data visualization through JTables
- Automated profit/stock calculations and implemented a Low Stock Alert with transaction logging

Web Development Projects

WaveCoin - Decentralized Token Application

Motoko, Internet Computer (ICP), DFX

Engineered this to understand how blockchain transactions actually work under the hood, beyond just reading about crypto.

- Engineered custom cryptocurrency logic for minting and transferring tokens on the blockchain
- Deployed canisters and integrated a frontend for real-time balance tracking

Advanced React Blog Platform

React.js, Node.js, Express, PostgreSQL/MongoDB

Architected this as a deeper dive into real authentication and session handling in a full-stack app, beyond basic CRUD.

- Developed a complete CRUD application with secure user authentication and session management

Additional Projects

Also built an AI Blog Platform with AI content generation and a RAG chatbot, using React and the Groq LLaMA API. There's also an E-Commerce Console App in Java and MySQL that simulates a full shopping and wallet system. And PeakNFT, an NFT minting platform built with React.js and Solidity/Motoko, using decentralized storage for metadata. Full code for all of these is on GitHub at <https://github.com/Upendrasai1101>